📓 MindAssess Platform Overview

📌 Overview

This is a comprehensive documentation of the MindAssess platform. The project is a full-stack MERN (MongoDB, Express, React, Node) application designed for clinical-grade mental health screening with a premium, responsive UI.

🎨 Frontend Architecture (/frontend)

The frontend is built on React 19 and Vite, utilizing TailwindCSS for styling and Framer Motion for high-end animations. It follows a Feature-Based Architecture, where code is grouped by domain logic rather than just type.

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1. Root Configuration Files

index.html

The entry window. Defines the site title, your custom X (logo) favicon/tab-icon, and the #root mounting point for React.

App.jsx

The central router.

- Logic: Uses react-router-dom to manage all page transitions.
- Context Wrapping: Wraps the entire site in AuthProvider (for login state) and ThemeProvider (for dark mode).
- Routing: Defines paths like /assessment, /dashboard, and /health-library/*.

main.jsx

The bootstrapper. Initializes the React root and imports index.css.

index.css

Implements the global design system, including font family (Inter/Sans), glassmorphism classes, and core color tokens.

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2. Core Components (/src/components)

This folder contains UI elements reused across multiple features.

Layout/

Navbar.jsx

Contains the sticky navigation, the "Health Library" Mega-menu trigger, and the mobile hamburger drawer.

Footer.jsx

Global footer containing social links (the new X icon), quick navigation, and the iCall Helpline crisis card.

UI/

Antigravity.jsx

A complex 3D particle background using React Three Fiber. It creates an interactive, magnetic field effect linked to mouse movement.

HealthLibraryDropdown.jsx

The desktop mega-menu that appears when hovering/clicking "Health Library" in the Navbar.

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routing/

ProtectedRoute.jsx

A logic wrapper that prevents users from accessing the dashboard or assessment if they are not logged in, redirecting them to /auth.

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3. Domains/Features (/src/features)

The "brain" of the app, split into focused modules.

assessment/

AssessmentEngine.jsx

The most complex component. It handles the 45+ questions, manages state transitions (sliding animations), and sends the finalized responses to the backend for scoring.

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dashboard/

UserDashboard.jsx

Fetching and displaying the user's history using Axios.

- Logic: It calculates longitudinal markers and conditionally reveals the "Healthy Lifestyle" section only after the first test is completed.

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health-library/

DiseasesPage.jsx

An encyclopedia of mental health disorders with a sticky navigation sidebar for easy scrolling.

SupplementsPage.jsx

An herb/vitamin guide using Accordion designs to show/hide medical details.

LifestylePage.jsx

A media-rich hub with YouTube therapeutic sessions (Yoga, Depression recovery, Overthinking guides).

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landing/

LandingPage.jsx

The high-converting homepage featuring the hero section, feature spotlights, and trust markers.

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4. Shared State Logic (/src/context)

AuthContext.jsx

Manages global user state (Current User, Token, Loading status). It handles persistent login by checking localStorage on page refresh.

⚙️ Backend Architecture (/backend)

The backend is a robust Node.js/Express server that enforces security and manages the clinical data processing.

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1. Routing & Logic Flow

server.js

The main entry point.

- Initializes MongoDB Connection
- Registers middleware (CORS, JSON parsing)
- Attaches route handlers

routes/

Defines the API endpoints (e.g., /api/auth/register, /api/assessment/history).

controllers/

Contains the heavy logic.

- assessmentController.js: Calculates the clinical magnitude of responses and generates the final burden scores.
- authController.js: Handles account creation and verifies passwords using Bycrypt.

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2. Data Persistence

models/

Defines the data shapes (Schemas) for MongoDB.

- User.js: Stores email, hashed passwords, and profile details.
- Assessment.js: Stores the raw test results, scores, and timestamps.

middleware/

authMiddleware.js

Shields the database. It validates JWT (JSON Web Tokens) to ensure only the owner can see their mental health history.

🛠 How it all works together

Authentication

A user registers (AuthPage), the backend creates a unique User ID and returns a Token.

Assessment

The user takes a test (AssessmentEngine). The engine uses local state to keep track of answers and then transmits an array of results to the backend.

Calculation

The backend (assessmentController) applies weighted algorithms to the data and saves it to the database.

Reporting

The Dashboard fetches this record (Axios), visualizes the scores via the UI, and unlocks specific "Lifestyle" recommendations based on that history.

Branding

Every interaction is smoothed over by the Antigravity background and Framer Motion animations to reduce user anxiety during clinical screening.

🔄 Routing System

The project uses React Router DOM v7 (the latest version of the industry standard for React navigation) to manage all URL routing and page transitions.

How it is implemented:

- Centralized Routing: All paths (like /dashboard, /assessment, and /health-library) are defined in the App.jsx file using the Routes and Route components.
- Navigation Components: Specifically, the Link component and the useNavigate hook are used throughout the app (like in the Navbar and Buttons) to change pages without reloading the browser.
- Protected Assets: I've implemented a ProtectedRoute wrapper that uses React Router logic to check if a user is authenticated before allowing them to access "private" pages like the assessment or the health dashboard.
- Scroll Management: The system is designed to reset the scroll position using useEffect when you switch routes, ensuring a smooth transition between different sections of the platform.

This ensures a modern "Single Page Application" (SPA) feel, where transitions happen instantly without heavy server requests.

🧠 Clinical Assessment System

The MindAssess platform captures data across 5 Core Clinical Categories. Each category uses a "Weight-Matrix" to determine the physiological and psychological burden.

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1. The 5 Data Capture Categories

Category	What it Measures
Anxiety Markers	Nervous system regulation, panic triggers, and uncontrollable worry
Depression Markers	Emotional affect, hopelessness, energy levels, and anhedonia
Focus & ADHD	Executive functioning, impulse control, and concentration stability
Trauma & Stress	Flashbacks, intrusive thoughts, and emotional detachment levels
Impact & Function	Sleep quality, social avoidance, and occupational productivity

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2. The Weighting System (Options)

We use a Linear Weighting Matrix for every response. Unlike simple quizzes, our system considers "Disagreeing" as a positive clinical indicator that lowers your total burden score.

- Strongly Agree (+2): High Symptom expression
- Agree (+1): Moderate Symptom presence
- Disagree (-1): No significant symptom detected
- Strongly Disagree (-2): Strong clinical resilience (High regulation)

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3. Demo Test Example (Simulation)

Questions & Responses

Question	User Response	Weight
"I find it hard to control my worrying."	Strongly Agree	+2
"I experience sudden surges of panic."	Agree	+1
"I feel on edge or restless."	Disagree	-1

Calculation Flow

Category Score:
(+2) + (+1) + (-1) = 2

Threshold Validation

The backend looks at the score 2.
According to our determine Levellogic, a score of 2 is the boundary for "Mildly Stressed."

Result Generation

The system identifies that you have worry control issues (the +2 weight) but are not physically restless (-1).

It generates a report card showing a **33% Burden Bar (2 out of max 6)** for Anxiety, with the insight:
"Minor physiological hyper-arousal detected. Short breathing exercises may help during peaks."

Final Global Scoring

The Global Total Score is simply the sum of all 15 questions across all categories.

- Maximum Possible Score: +30 (Critical Clinical Burden)
- Minimum Possible Score: -30 (Exceptional Psychological Stability)

User Example:
If a user finishes the demo with a Global Total of 12, they fall into the "Moderate Burden" bracket and are advised to seek professional consultation.